

1. The number of air molecules per  $\text{cm}^3$  is increased from  $3 \times 10^{19}$  to  $12 \times 10^{19}$ . The ratio of collision frequency of air molecules before and after the increase in number respectively is :

[2023 (06 Apr Shift 1)]

- (1) 0.75  
(2) 1.25  
(3) 0.50  
(4) 0.25

2. The temperature of an ideal gas is increased from 200 K to 800 K. If r.m.s. speed of gas at 200 K is  $v_0$ . Then, r.m.s. speed of the gas at 800 K will be:

[2023 (06 Apr Shift 2)]

- (1)  $\frac{v_0}{4}$   
(2)  $v_0$   
(3)  $4v_0$   
(4)  $2v_0$

3. The temperature at which the kinetic energy of oxygen molecules becomes double than its value at  $27^\circ\text{C}$  is

[2023 (08 Apr Shift 2)]

- (1)  $927^\circ\text{C}$   
(2)  $327^\circ\text{C}$   
(3)  $1227^\circ\text{C}$   
(4)  $627^\circ\text{C}$

4. Match List I with List II:

|     | List I                                                                         |       | List II                 |
|-----|--------------------------------------------------------------------------------|-------|-------------------------|
| (A) | 3 Translational degrees of freedom                                             | (I)   | Monoatomic gases        |
| (B) | 3 Translational, 2 rotational degrees of freedoms                              | (II)  | Polyatomic gases        |
| (C) | 3 Translational, 2 rotational and 1 vibrational degrees of freedom             | (III) | Rigid diatomic gases    |
| (D) | 3 Translational, 3 rotational and more than one vibrational degrees of freedom | (IV)  | Nonrigid diatomic gases |

Choose the correct answer from the options given below:

[2023 (10 Apr Shift 1)]

- (1) (A)–(I), (B)–(III), (C)–(IV), (D)–(II)  
(2) (A)–(IV), (B)–(III), (C)–(II), (D)–(I)  
(3) (A)–(IV), (B)–(II), (C)–(I), (D)–(III)  
(4) (A)–(I), (B)–(IV), (C)–(III), (D)–(II)

5. A gas mixture consists of 2 moles of oxygen and 4 moles of neon at temperature  $T$ . Neglecting all vibrational modes, the total internal energy of the system will be:

[2023 (10 Apr Shift 2)]

- (1)  $11RT$   
(2)  $8RT$   
(3)  $4RT$   
(4)  $16RT$

6. Three vessels of equal volume contain gases at the same temperature and pressure. The first vessel contains neon (monoatomic), the second contains chlorine (diatomic) and third contains uranium hexafluoride (polyatomic). Arrange these on the basis of their root mean square speed ( $v_{rms}$ ) and choose the correct answer from the options given below:

[2023 (11 Apr Shift 1)]

- (1)  $v_{rms}(\text{mono}) > v_{rms}(\text{dia}) > v_{rms}(\text{poly})$   
(2)  $v_{rms}(\text{mono}) = v_{rms}(\text{dia}) = v_{rms}(\text{poly})$   
(3)  $v_{rms}(\text{mono}) < v_{rms}(\text{dia}) < v_{rms}(\text{poly})$   
(4)  $v_{rms}(\text{dia}) < v_{rms}(\text{poly}) < v_{rms}(\text{mono})$

7. The root mean square speed of molecules of nitrogen gas at  $27^\circ\text{C}$  is approximately: (Given mass of a nitrogen molecule  $= 4.6 \times 10^{-26}$  kg and take Boltzmann constant  $k_B = 1.4 \times 10^{-23} \text{ J K}^{-1}$ )  
[2023 (11 Apr Shift 2)]  
(1)  $27.4 \text{ m s}^{-1}$   
(2)  $91 \text{ m s}^{-1}$   
(3)  $1260 \text{ m s}^{-1}$   
(4)  $523 \text{ m s}^{-1}$
8. If the r.m.s speed of chlorine molecule is  $490 \text{ m s}^{-1}$  at  $27^\circ\text{C}$ , the r.m.s speed of argon molecules at the same temperature will be (Atomic mass of argon  $= 39.9 \text{ u}$ , molecular mass of chlorine  $= 70.9 \text{ u}$ )  
[2023 (12 Apr Shift 1)]  
(1)  $551.7 \text{ m s}^{-1}$   
(2)  $651.7 \text{ m s}^{-1}$   
(3)  $451.7 \text{ m s}^{-1}$   
(4)  $751.7 \text{ m s}^{-1}$
9. The rms speed of oxygen molecule in a vessel at particular temperature is  $\left(1 + \frac{5}{x}\right)^{\frac{1}{2}} v$ , when  $v$  is the average speed of the molecule. The value of  $x$  will be: (take  $\pi = \frac{22}{7}$ )  
[2023 (13 Apr Shift 1)]  
(1) 27  
(2) 8  
(3) 28  
(4) 4
10. The mean free path of molecules of a certain gas at STP is  $1500d$ , where  $d$  is the diameter of the gas molecules. While maintaining the standard pressure, the mean free path of the molecules at  $373 \text{ K}$  is approximately:  
[2023 (13 Apr Shift 2)]  
(1)  $750d$   
(2)  $1098d$   
(3)  $2049d$   
(4)  $1500d$
11. A flask contains Hydrogen and Argon in the ratio 2 : 1 by mass. The temperature of the mixture is  $30^\circ\text{C}$ . The ratio of average kinetic energy per molecule of the two gases  $\left(\frac{K_{\text{argon}}}{K_{\text{hydrogen}}}\right)$  is: (Given : Atomic Weight of  $\text{Ar} = 39.9$ )  
[2023 (15 Apr Shift 1)]  
(1) 2  
(2) 1  
(3) 39.9  
(4)  $\frac{39.9}{2}$

## ANSWER KEYS

1. (4)      2. (4)      3. (2)      4. (1)      5. (1)      6. (1)      7. (4)      8. (2)  
9. (3)      10. (3)      11. (2)

1. (4)

The given data is

$$n_1 = 3 \times 10^{19}$$

$$n_2 = 12 \times 10^{19}$$

The mean time between two collision has the meaning of a period  $\tau$ , since it indicates the repetitive time intervals in which on average collisions take place. The reciprocal of the time  $\tau$  can be understood as the collision frequency  $f$ , which indicates the number of collisions per unit time. The collision frequency is also directly proportional to the number molecules per unit volume.

$$f \propto \frac{1}{\tau}$$

$$\Rightarrow f \propto n$$

$$\Rightarrow \frac{f_1}{f_2} = \frac{n_1}{n_2} = \frac{3 \times 10^{19}}{12 \times 10^{19}} = 0.25$$

2. (4)

At 200 K, the rms speed is given by

$$v_0 = \sqrt{\frac{3R \times 200}{M}} \dots (1)$$

At 800 K, the rms speed is given by

$$v' = \sqrt{\frac{3R \times 800}{M}} \dots (2)$$

Divide equation (2) by equation (1) and solve to calculate the required speed.

$$\begin{aligned} \frac{v'}{v_0} &= \frac{\sqrt{\frac{3R \times 800}{M}}}{\sqrt{\frac{3R \times 200}{M}}} \\ &= \sqrt{\frac{800}{200}} \\ &= 2 \end{aligned}$$

$$\Rightarrow v' = 2v_0$$

3. (2)

The kinetic energy is given by  $KE = \frac{f}{2} kT$ .

Let the kinetic energy be  $K_1$  at  $27^\circ\text{C}$  and the kinetic energy be  $K'$  at  $T'$  temperature. As given in the question,  $K' = 2K_1$ . Hence,

$$\frac{K'}{K_1} = \frac{T'}{T}$$

$$\Rightarrow 2 = \frac{T'}{300}$$

$$\Rightarrow T' = 600 \text{ K}$$

In Celsius the temperature is  $T' - 273 = 327^\circ\text{C}$ .

4. (1)

The formula for degree of freedom is given by

$$f = 3N - k$$

where  $N$  is the number of particles,  $k$  is the individual relationship in the system.

For a monoatomic gas particle,  $N = 1$ ,  $k = 0$ . So,  $f = 3$

Hence, monoatomic gas has 3 translational degree of freedom.

For a rigid diatomic gas molecule, there are 5 degrees of freedom, 3 translational and 2 rotational. A diatomic gas can have two extra degrees of freedom due to rotation along two independent axes.

A non rigid diatomic molecule has one extra degree of freedom than a rigid one. This is the vibrational degree of freedom. Hence, it has 6 degrees of freedom.

So, a non rigid diatomic gas has 3 translational, 2 rotational and 1 vibrational degrees of freedom.

A polyatomic gas has more than two atoms per molecule. Hence, the total number of degrees of freedom is 3 translational, 3 rotational and more than one vibrational degree of freedom.

5. (1)

The internal energy of a gas is given by  $U = nC_V T$ .

A diatomic gas has 3 translational and 2 rotational degrees of freedom. A monoatomic gas has 3 translational degrees of freedom.

Hence, the internal energy of the mixture =  $n_1 C_{V1} T + n_2 C_{V2} T$

$$= 2 \times \left(\frac{5R}{2}\right) T + 4 \times \frac{3R}{2} T$$

$$= 5RT + 6RT$$

$$= 11RT$$

6. (1)

The RMS speed is given  $v_{rms} = \sqrt{\frac{3RT}{M}}$ .

The temperature and pressure is same for all the gases. So the speed is inversely proportional to the square root of the molar mass of the gases.

$$v_{rms}(\text{mono}) = \sqrt{\frac{3RT}{4 \times 10^{-3}}}$$

$$v_{rms}(\text{dia}) = \sqrt{\frac{3RT}{71 \times 10^{-3}}}$$

$$v_{rms}(\text{poly}) = \sqrt{\frac{3RT}{352 \times 10^{-3}}}$$

$$v_{rms} \propto \frac{1}{\sqrt{M}}$$

Hence,

$$v_{rms}(\text{poly}) < v_{rms}(\text{dia}) < v_{rms}(\text{mono}).$$

7. (4)

The RMS value of a gas is given by

$$v_{rms} = \sqrt{\frac{3RT}{M}}$$

$$= \sqrt{\frac{3k_B T}{m}}$$

Substituting the given values,

$$v_{rms} = \sqrt{\frac{3 \times 1.4 \times 10^{-23} \times 300}{4.6 \times 10^{-26}}}$$

$$\approx 523 \text{ m s}^{-1}$$

8. (2)

The formula to calculate the rms speed of the argon molecules is given by

$$v_{Ar} = \sqrt{\frac{3RT}{M_{Ar}}} \dots (1)$$

The formula to calculate the rms speed of chlorine molecule is given by

$$v_{Cl} = \sqrt{\frac{3RT}{M_{Cl}}} \dots (2)$$

Divide equation (1) by equation (2) and simplify to obtain the required ratio.

$$\frac{v_{Ar}}{v_{Cl}} = \sqrt{\frac{M_{Cl}}{M_{Ar}}} \dots (3)$$

Substitute the values of the known parameters into equation (3) and solve to calculate the required value.

$$\frac{v_{Ar}}{490 \text{ m s}^{-1}} = \sqrt{\frac{71}{40}}$$

$$\Rightarrow v_{Ar} = \sqrt{\frac{71}{40}} \times 490 \text{ m s}^{-1}$$

$$\approx 651.7 \text{ m s}^{-1}$$

9. (3)

The RMS speed is given by

$$v_{\text{rms}} = \sqrt{\frac{3RT}{M}}$$

The average speed is given by

$$v_{\text{avg}} = \sqrt{\frac{8RT}{\pi M}}$$

The ratio is

$$\Rightarrow \frac{v_{\text{rms}}}{v_{\text{avg}}} = \sqrt{\frac{\frac{3RT}{M}}{\frac{8RT}{\pi M}}} = \sqrt{\frac{3\pi}{8}} = \sqrt{\frac{3}{8} \times \frac{22}{7}}$$

$$\Rightarrow \frac{v_{\text{rms}}}{v_{\text{avg}}} = \sqrt{\frac{33}{28}}$$

Comparing it with the given value  $\left(1 + \frac{5}{x}\right)^{\frac{1}{2}}$ ,

$$\frac{v_{\text{rms}}}{v_{\text{avg}}} = \sqrt{1 + \frac{5}{28}}$$

$$\Rightarrow x = 28$$

10. (3)

The mean free path is given by the relation,

$$\lambda = \frac{RT}{\sqrt{2} \pi d^2 N_A P}$$

So,  $\lambda \propto T$

Let  $\lambda'$  be the new mean free path.

Hence,

$$\frac{1500d}{\lambda'} = \frac{273}{373}$$

$$\Rightarrow \lambda' = 2049.45d \approx 2049d$$

11. (2)

The formula to calculate the average kinetic energy per molecule of a gas at any temperature  $T$  is given by

$$K_{\text{avg}} = \frac{3}{2} k_B T$$

where,  $k_B$  is Boltzmann constant.

From above equation, it can be concluded that the average kinetic energy per molecule is independent of the molecular weight of the gas.

Hence, the required ratio is 1 : 1.